

A CONTEMPORARY APPROACH TO RESEARCH AND STATISTICS IN PSYCHOLOGY



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A Contemporary Approach to Research and Statistics in Psychology

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This textbook represents a labour of love and a deep commitment to my students.

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This book is dedicated to Michelle, who took four years of research methods and statistics in her psychology degree, and received distinctions and high distinctions in these subjects but admitted that she did not learn anything. She only knew how to click the right buttons in SPSS. I also dedicate this book to my niece and nephews, and I hope that they will develop the same interest and curiosity towards numbers and research.

Acknowledgement of Country

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James Cook University is committed to building strong and mutually beneficial partnerships that work towards closing the employment, health and education gap for Australian Aboriginal and Torres Strait Islander peoples. Our students come from many backgrounds, promoting a rich cultural and experiential diversity on campus. We acknowledge the Aboriginal and Torres Strait Islander peoples as the Traditional Custodians of the Australian lands and waters where our staff and students live, learn and work. We honour the unique cultural and spiritual relationship to the land, waters and seas of First Australian peoples and their continuing and rich contribution to James Cook University and Australian society and we recognise that these lands have always been places of teaching and learning. We also pay respect to ancestors and Elders past and present.



Kassandra Savage (JCU Alumni), 'Coming Together and Respecting Difference', acrylic on canvas, 2014, 90cm x 90cm. © Kassandra Savage, reproduced with permission of the artist

About the Book



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About the Author

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Dr Klaire Somoray (they/she) is a Lecturer in Psychology at the College of Healthcare Sciences, James Cook University. They completed their PhD at Queensland University of Technology, within the Centre for Accident Research and Road Safety Queensland (CARRS-Q).

Dr Somoray has a diverse research portfolio. Due to their primary expertise in statistical modelling and quantitative research methodologies, they have been involved in different research projects, including research on traffic psychology, workplace health, safety and wellbeing, and criminology.

Currently, they have a strong interest in examining the positive and negative impacts of data and technology at the individual- and societal-level.

If you would like to provide feedback on this OER, please do so by contacting Klaire by email: klaire.somoray@jcu.edu.au

Why do we Need Another Book About Research and Statistics?

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Psychology is currently facing a crisis. In 2015, a study published by the Open Science Collaboration found that only 36% of 100 experimental and correlational studies from three top-ranking journals could be replicated.^[1] This news was widely covered by media outlets around the world, including the *New York Times* and the *Atlantic*. As a wide-eyed (and very naive) undergraduate psychology student at that time, who was eager to cure humanity of its mental health issues, this was not what I had signed up for.

While some researchers argue that psychology is not in crisis,^[2] it is still important to understand the reasons behind the replicability crisis and how the findings of the Open Science Collaboration study can be used to promote better scientific practices. This Open Education Resource is a humble attempt to assist future psychology students in conducting research that is robust and reliable.

It is also my hope that this resource will help ease the burden of mandatory research and statistics coursework for students who aspire to make a positive impact on people's lives through the field of psychology.

My Experience with Statistics

My relationship with statistics is unique compared to that of my peers. Many people are surprised or confused when I tell them that I love statistics. Sometimes, I am even surprised by this myself. If someone had told my 16-year-old self that I would one day be teaching statistics at the undergraduate level, I would have thought that my 30-something self had lost their mind.

I do admit that I have always loved puzzles, and I was fortunate enough to have passionate educators who taught me statistics in a fun and non-threatening manner, using resources such as Andy Field's book.^[3] I viewed statistics class as another puzzle to be solved, albeit a very challenging one.

Because of this, I developed a deep curiosity about statistics and began to see it in a different light. I realised that maybe statistics isn't as difficult as it seems, or maybe, a lot of educators (even myself) make it more difficult than it needs to be. Maybe, just *maybe*, there is a way to make it more intuitive for students.

Dustin Fife, who inspired me to write this Open Education Resource, even suggested renaming statistics as "simplistics" to make it more approachable.^[4]

The Philosophy Behind this Little Book

When I first started teaching undergraduate statistics, it was challenging to find a textbook that aligns with my perspective on statistics. I was also teaching intensive classes. While I loved Andy Field's textbook, I couldn't use it as a prescribed text because it required a lot of reading through stories to get to the statistical knowledge.^[5] Additionally, it didn't cover certain topics, such as a brief introduction to research methods, open science principles and critiquing research. I ultimately felt that my students would be better served by a book that closely follows my philosophy of teaching research methods and statistics in psychology.

The statistics section of this book will be similar to Field's book, but the content is substantially less comprehensive and less engaging.

The goal of this book is to help future scholars of psychology conduct better science by teaching statistics in an intuitive manner. I also hope that my students will not only learn a thing or two about research and statistics, but also find it useful in their everyday life and future practice. Perhaps some students may even be inspired to become statisticians or data analysts in the future!

-
1. Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251). doi.org/10.1126/science.aac4716 ↩
 2. Barrett, L. F. (2015). Psychology is not in crisis. *The New York Times*. www.nytimes.com/2015/09/01/opinion/psychology-is-not-in-crisis.html ↩
 3. Field, A. (2013). *Discovering statistics using IBM SPSS statistics*. Sage. ↩
 4. quantpsych.net/stats_modeling/ ↩
 5. If you think this approach will work for you, please purchase a copy of Andy Field's textbook - it is very entertaining! ↩

CHAPTER OVERVIEW

Chapter 1: Research and Statistical Thinking in Everyday Life

Learning Objectives

After reading this chapter, you should be able to:

- discuss research and statistical thinking in everyday life
- describe the central goals and fundamental concepts of statistics.

We are drowning in information, but we are starved for knowledge – John Naisbitt^[1]

When was the last time you made a decision? Perhaps you decided to sleep in this morning instead of going for a run, or maybe you chose to make coffee when you woke up. Even reading this book is a decision you've made.

As you made these decisions, what thoughts were going through your mind? Maybe when you decided to sleep in, you rationalised that getting more sleep is better for you than exercising. Perhaps you decided to make coffee because it gives you a caffeine boost and helps you focus on the task at hand. And maybe you're reading this book because you want to see if my approach will help you get a better grade on your next statistics assignment.

What if I told you that statistical thinking can help you make better decisions? That may sound like a pyramid scheme, but I believe this sentiment is true.

Many people don't realise that they already use statistical thinking on a daily basis. For example, students are often adept at this when they calculate how many points they need on the next assessment to pass the overall subject.

A Note on Dealing with Anxiety Related to Statistics

Before we continue, I want to acknowledge the anxiety that many people feel when they first take a statistics class. The use of unfamiliar statistical software can also feel overwhelming. However, feeling anxious does not mean that you are bad at statistics or that you don't understand it. Statistics are not necessarily intuitive.

Anxiety can be uncomfortable, but psychology tells us that this kind of emotional arousal can actually help us perform better on tasks by focusing our attention. So, if you start to feel anxious about the materials in your statistics course, remind yourself that other students in your class are likely feeling the same way and that the arousal could actually help you perform better, even if it doesn't seem like it.

Before we delve into the details, let's first discuss what statistics can do for us.

[1.1: What can Statistics do for us?](#)

[1.2: If I Apply Statistical Thinking all the Time then why do I find it Difficult?](#)

[1.3: Big Ideas in Statistics](#)

[1.4: Data is/are](#)

1. Naisbitt, J. (1982). *Megatrends: Ten new directions transforming our lives*. Warner Books. [↩](#)

1.1: What can Statistics do for us?

There are four main things we can do with statistics, which we already do on a regular basis:

- **Describe:** The world is complex, and we often need to simplify it to understand it.
- **Explain:** We can use data to explain a phenomenon.
- **Decide:** We also often need to make decisions based on data, usually in the face of uncertainty.
- **Predict:** We often wish to make predictions about new situations based on our knowledge of previous situations.

Statistical thinking is a tool that can help us achieve these goals. As you read through the points above, you may have noticed that we already do these behaviours in our everyday lives. Every day, we describe things (e.g., the weather is not great today), explain things (e.g., the food was salty because I put too much salt while cooking it), make decisions (e.g., I will take this route because it will be shorter), and predict things (e.g., the food in this restaurant is going to be great because it's been recommended to me by many people). The only difference with statistical thinking is that we use data as evidence for these everyday behaviours.

Let's Look at an Example

To illustrate this point, let's take a common question that many people are interested in: How do we decide what's healthy to eat? There are many different sources of guidance, from government dietary guidelines to diet books and different bloggers.

Now, let's focus on a specific question: *Is saturated fat in our diet a bad thing?*

One way to answer this question is through common sense. If we eat fat, it's going to turn into fat in our bodies, right? And we've all seen photos of arteries clogged with fat, so eating fat is going to clog our arteries, right?

Another way to answer this question is by listening to authority figures. For example, the third point under the Australian Dietary Guidelines states: "Limit intake of foods high in saturated fat such as many biscuits, cakes, pastries, pies, processed meats, commercial burgers, pizza, fried foods, potato chips, crisps and other savoury snacks."^{[1][2]}

Finally, we might look at actual scientific research. Let's start by looking at a large study called the long-running **Prospective Urban and Rural Epidemiological** study aka the **PURE** study, which has examined diets and health outcomes (including death) in more than 135,000 people from 18 different countries. In one of the analyses of this dataset (published in *The Lancet* in 2017 by Dehghan et al., 2017),^[3] the PURE investigators reported an analysis of how intake of various classes of macronutrients (including saturated fats and carbohydrates) was related to the likelihood of dying during the time that people were followed. People were followed for a median of 7.4 years.

Figure 1.1.1 below plots some of the data from the study (extracted from the paper), showing the relationship between the intake of both saturated fats and carbohydrates and the risk of dying from any cause.

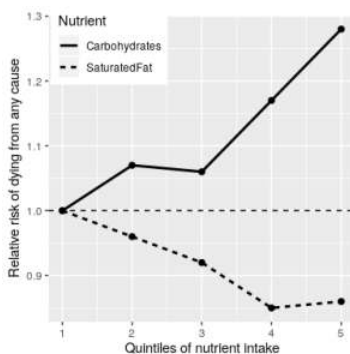


Figure 1.1.1 "A plot of data from the PURE study, showing the relationship between death from any cause and the relative intake of saturated fats and carbohydrates" by [Russell A. Poldrake](#) is licensed under CC BY-NC 4.0

This plot is based on 10 numbers. To obtain these numbers, the researchers split the group of 135,335 study participants (which we call the "sample") into five groups ("quintiles") after ordering them in terms of their intake of either of the nutrients; the first quintile contains the 20% of people with the lowest intake, and the 5th quintile contains the 20% with the highest intake. The researchers then computed how often people in each of those groups died during the time they were being followed. The figure expresses this in terms of the *relative risk* of dying in comparison to the lowest quintile: If this number is greater than 1 it means that people in that group are *more* likely to die than the people in the lowest quintile, whereas if it's less than one it means that people in that group are *less* likely to die than the people in the lowest quintile.

The figure is pretty clear: People who ate more saturated fat were *less* likely to die during the study, with the lowest death rate seen for people who were in the fourth quintile (that is, who ate more fat than the lowest 60% but less than the top 20%). The opposite is seen for carbohydrates; the more carbs a person ate, the more likely they were to die during the study. This example shows how we can use statistics to *describe* a complex dataset in terms of a much simpler set of numbers; if we had to look at the data from each of the study participants at the same time, we would be overloaded with data and it would be hard to see the pattern that emerges when they are described more simply.

The numbers in Figure 1.1.1 seem to show that *deaths decrease with saturated fat and increase with carbohydrate intake*, but we also know that there is a lot of uncertainty in the data; there are some people who died early even though they ate a low-carb diet, and, similarly, some people who ate a ton of carbs but lived to a ripe old age. Given this variability, we want to *decide* whether the relationships that we see in the data are **large enough** that we wouldn't expect them to occur randomly if there was not truly a relationship between diet and longevity.

Statistics provide us with the tools to make these kinds of decisions, and often people from the outside view this as *the* main purpose of statistics. But as we will see throughout the book, this need for black-and-white decisions based on fuzzy evidence has often led researchers astray.

We can also make predictions about future outcomes based on the available data that we have. For example, a life insurance company might want to use data about a particular person's intake of fat and carbohydrate to predict how long they are likely to live. An important aspect of prediction is that it requires us to generalise from the data we already have to some other situation, often in the future. However, if our conclusions were limited to the specific people in the study at a particular time, then the study would not be very useful. In general, researchers must assume that their particular sample is representative of a larger *population*, which requires that they obtain the sample in a way that provides an unbiased picture of the population. For example, if the PURE study had recruited all of its participants from religious sects that practice vegetarianism, then we probably wouldn't want to generalise the results to people who follow different dietary standards.

Chapter attribution

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1. You can read the guidelines here: www.eatforhealth.gov.au/sites/default/files/2022-09/n55a_australian_dietary_guidelines_summary_131014_1.pdf ↩
2. You might hope that these guidelines would be based on good science, and in some cases they are, but as Nina Teicholz outlined in her book *Big Fat Surprise* (Teicholz 2014), this particular recommendation seems to be based more on the dogma of nutrition researchers than on actual evidence. ↩
3. Dehghan, M., Mente, A., Zhang, X., Swaminathan, S., Li, W., Mohan, V., ... Garcia, R. (2017). Associations of fats and carbohydrate intake with cardiovascular disease and mortality in 18 countries from five continents (PURE): A prospective cohort study. *The Lancet*, 390(10107), 2050-2062. [https://doi.org/10.1016/S0140-6736\(17\)32252-3](https://doi.org/10.1016/S0140-6736(17)32252-3) ↩

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